Safari Research Proposal

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Group Name

Safari

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Research Project Title

Attention Allocation and Navigation Information Design: for Newcomers to Hong Kong

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Research Background

Our starting point is the group's experience of arriving in an unfamiliar place: a trip intended for leisure, exploration or an everyday errand can become dominated by the work of getting there. Reading a route, matching it to the street, locating an exit or entrance and repeatedly checking progress may leave little attention for the surroundings. We are concerned that some newcomers may feel overwhelmed, enjoy the outing less, lose the desire to continue exploring, or abandon the intended destination. These are motivating observations and possible consequences to investigate, not established findings about Hong Kong newcomers.

Qualitative evidence gives this concern a basis. Schmitt et al. (2019) describe unfamiliar transit journeys as involving sustained vigilance, active information gathering and uncertainty, in contrast to the opportunities for relaxation on familiar trips. Participants also described navigation mistakes despite preparation. Although this is transit research, it suggests why reaching a destination can itself become a demanding activity. For this project, feeling overwhelmed means a participant's experience of being unable to comfortably keep up with the information and decisions required at a particular moment. It is an experiential description, not a diagnosis or a synonym for objectively measured cognitive overload.

Unfamiliarity turns ordinary movement into repeated decisions

Without familiar reference points, a person has to establish where they are, relate instructions to visible surroundings and decide whether the next action is correct. A map can provide directions while leaving uncertainty about how to interpret them on the ground. Research with visitors to an unfamiliar city found that digital navigation did not eliminate anxiety relative to paper maps or local signage; following a suggested route also did not guarantee awareness of the surrounding space (Vaez et al., 2020). This suggests that information availability and confidence in acting are different design concerns.

Navigation competes with attending to the city

A person may need to read a display, watch pedestrians and traffic, identify a landmark and remember the next instruction almost simultaneously. Multiple resource theory explains how concurrent tasks can interfere when they rely on overlapping processing resources (Wickens, 2008). Research on mobile interaction describes attention fragmented by urban situations (Oulasvirta et al., 2005), while situational impairments show how environmental and bodily conditions can constrain technology use (Wobbrock, 2019). Applied to our question, these perspectives suggest that repeated shifts between screen and surroundings may make navigation feel demanding, even when each instruction is individually understandable. This is a theoretical interpretation to examine, not proof that every attention shift is harmful.

More information can help or become difficult to use

A crowded map, an instruction arriving at the wrong moment or a missing confirmation cue can create different problems. The issue may be selecting relevant information, interpreting it or deciding whether it can be trusted. Virtual navigation studies link landmark display conditions to cognitive demands and show that reducing task-irrelevant map content is not uniformly beneficial across contextual demands (Cheng et al., 2023; Bartling et al., 2026). Consequently, feeling overwhelmed may reflect a mismatch between information, timing and task, rather than simply too much information. In real-world navigation, phone checks are associated with stops and route errors, but the causal direction is unresolved (Shimokihara et al., 2025). Repeated checking could therefore indicate a difficulty or an attempt to regain certainty.

Uncertainty can change the experience and willingness to explore

Navigation has emotional as well as practical consequences. Perceived wayfinding difficulty is associated with anxiety among tourists in unfamiliar destinations (Chang, 2013). Qualitative park research shows that people value route-distance and orientation information when judging whether they feel confident enough to venture further (Hardwicke et al., 2024). These findings make enjoyment, confidence and willingness to continue relevant outcomes, alongside arrival. They do not establish that navigation causes energy depletion and subsequent withdrawal from exploration among Hong Kong newcomers. That proposed connection remains a question for participants' accounts, including experiences in which unfamiliarity is enjoyable or additional information is reassuring.

We will therefore investigate how newcomers experience demanding moments during unfamiliar walking navigation, how those moments relate to their wider outing, and what information support they need. The contribution will be a contextual classification connecting information roles, presentation conditions, reported experience and action, followed by preliminary guidance on when and how to communicate navigation information. The study will explore an experience and its possible explanations rather than assume a universal causal sequence or a predetermined voice, haptic or visual solution.

Research Objectives

- Identify information needs and observable interaction episodes across an unfamiliar walking journey.
- Identify participant-reported moments of feeling overwhelmed, examine possible sources of difficulty, and explore perceived consequences for enjoyment, confidence, continued exploration and reaching the destination.
- Develop a classification linking information roles, contextual demands and participant explanations, including contrasting cases.
- Derive evidence-linked guidance for presentation timing, detail and delivery, with clear limits on what has been evaluated.

Research Questions

Main question: How can navigation information be classified by its associated interaction burdens to inform interaction design methods for improving when and how it is presented and delivered?

The main question is retained from the group's registered topic. We interpret burden as contextual, rather than a fixed property of an information category.

- RQ1: What information do newcomers need at different stages of walking navigation, and how do they coordinate phone use with environmental attention?
- RQ2: When do participants report feeling overwhelmed or supported, what information and contextual conditions accompany these experiences, and how do they describe the consequences for their outing?
- RQ3: How can these findings inform a classification and design guidance concerning information timing, detail and delivery?

Data Collection Methods

Research approach and scope

We propose a focused qualitative field study combining walking observation, brief voluntary in-the-moment comments, video of key navigation episodes and a retrospective semi-structured interview. Behaviour, environment and participant accounts will be interpreted together. Descriptive event counts support analysis; the study does not directly measure neural attention, physiological fatigue or a causal effect of navigation on enjoyment.

The analysis unit is an information-interaction episode linking a trigger, information, action, reported experience and outcome. We distinguish participant-reported overwhelm from a researcher's suspicion of difficulty. A participant can report overwhelm during the walk or on reviewing an episode afterwards; visible hesitation, repeated checks or a wrong turn alone will never establish this experience. Episodes may also involve manageable effort, uncertainty without overwhelm, reassurance or smooth progress. Self-described mental effort and physical tiredness will be recorded separately.

RQ	Evidence needed	Collection and analysis
RQ1	Information needed, visible actions and journey stages	A short profile interview, observed route and episode timeline.
RQ2	Reported experience, contextual triggers and perceived outing consequences	Optional brief comments, key-episode video and retrospective interviewing; distinguish reports from researcher inference.
RQ3	Information roles, contextual demands and helpful support	Cross-case matrix linking reported experience, observed episodes and design propositions.

Preparation and pilot

Before participant research, we will prepare the protocol, information sheet, consent form and route plan for the required ethics review. Researchers will scout one public route, normally taking about 15–20 minutes, from a transport exit to a public destination entrance. A backup route will be used only when familiarity, access or safety makes the primary route unsuitable; route differences will be documented rather than treated as equivalent experimental conditions. One eligible pilot volunteer, excluded from the main sample, will test the procedure, media workflow and timing after clearance.

One integrated session per participant

Each main session will take approximately 45–60 minutes: 5–10 minutes for consent and a short profile interview; 15–20 minutes for the observed walk; and 25–30 minutes for the debrief, including selected video review. There will be no required second visit.

Participants will use their usual navigation application and settings at their own pace, without a speed target. One researcher will accompany them, note brief event markers and monitor the route; the second will operate the camera. Before walking, participants will be invited to say a short phrase, such as ‘mark this moment’, if anything feels difficult to manage or particularly helpful. They may briefly say what they are trying to work out, or remain silent and discuss it later. This is optional, episode-based verbalisation rather than continuous think-aloud. No prompt will ask them to feel or demonstrate overwhelm, and no research conversation will be requested while crossing roads. Researcher prompts and interruptions will themselves be logged.

At pre-scouted decision points, the camera researcher will record short contextual clips where permitted and separately consented; participant signals or emerging difficulties can also trigger a clip. Recording will cover visible signage, path choices, entrances, environmental conditions and, if agreed, relevant hand/device actions. Each clip will have a timestamp linked to an event note recording what preceded it, what was being sought, the participant’s exact words where available and what happened next. A small number of smooth or helpful episodes will also be recorded. Recording that starts after a difficulty cannot establish its onset: missing lead-in will be marked, not reconstructed. No participant will be asked to recreate distress for the camera.

The 25–30-minute debrief will begin with an open account, then review approximately three to five episodes using notes and selected short video extracts, including a smooth or helpful episode where available. We will ask ‘What were you trying to work out?’, ‘How did that moment feel?’ and ‘What made the next step easier or harder?’. Participants can confirm, reject or revise researcher interpretations. We will then ask whether the journey affected enjoyment, perceived remaining energy, confidence or desire to continue, with ‘no effect’ and positive experiences treated as equally valid. Within the same interview, one recent self-selected real outing can provide context for changing plans, continuing to explore or abandoning a destination; retrospective accounts and hypothetical intentions will remain separate from observed behaviour. Heat, walking effort, time pressure, mood and prior tiredness will be considered as alternative explanations.

Video review replaces the previous information-card activity to keep the session within 45–60 minutes; there is no additional visit. The pilot will test recording, timestamps, clip retrieval and the interview duration. Researchers will note whether the observed destination was reached independently, reached with help or not reached, and why the session ended. A researcher-imposed time limit or participant withdrawal will not automatically count as navigation failure. When the route cannot be completed within the planned session, the team will stop at a safe point and retain a brief debrief rather than pressure the participant to finish. Participants may decline video or audio and continue with notes-based observation and interviewing; capture mode will be recorded.

Analysis and synthesis

We will organise material in a cross-case qualitative matrix that retains links to the original notes, clips and interview accounts. Interview recordings will be transcribed and checked. Videos will receive a short event index; only relevant speech and selected episodes require detailed transcription or coding. After reviewing each complete case, two researchers will independently interpret the first two cases, discuss differences and refine a shared coding structure for the remaining material. Revisions and unresolved interpretations will be recorded.

The matrix will record participant, journey stage, information sought, presentation and environmental conditions, action, contemporaneous words, retrospective explanation, reported overwhelm status, response and outcome. It will distinguish experience reported during an event, experience recalled later, observed behaviour and researcher inference. Orientation, guidance, confirmation and recovery are provisional information roles, open to revision. Contrasting cases will include similar actions without reported overwhelm and demanding moments without a visible interruption. Multiple episodes from one person will not count as independent participants.

Design propositions will follow an explicit chain: observed episode → participant explanation → contextual interpretation → proposed information support. Each proposition will include a supporting example, a counterexample or boundary condition, and its evaluation status. We will review two anonymised cases against the emerging framework internally. This checks traceability, not effectiveness with new users. A prototype trial or repeat participant session is outside the core plan and would require a separate rationale, time allocation and any necessary ethics amendment.

Who the Participants Are

We will plan six core participants, with up to two additional participants only if evidence gaps and the time budget justify them, plus one separate pilot. Participants will be adults who moved to Hong Kong within the previous six months, use smartphone navigation, are unfamiliar with the selected route, and can comfortably undertake the walk. English or Mandarin will be supported. Time since arrival is an operational recruitment criterion, not a validated boundary of navigation competence. Route familiarity, tool experience and language familiarity will be recorded separately.

Recruitment will seek variation in navigation habits without subgroup quotas or a newcomer-versus-local comparison. Six core cases are a practical course-project target, not a claim of statistical representativeness or guaranteed saturation. After six sessions, we will review missing perspectives and weakly supported patterns before deciding whether up to two further cases are justified by the remaining time. If evidence is limited, the claims will be narrowed rather than treating the planned sample size as proof of adequacy.

How Participants Will Be Recruited

We will use purposive convenience recruitment through university and newcomer networks where invitations are permitted, with voluntary peer referrals. Interested people will contact the team privately. A short screener will check eligibility and route familiarity. We will avoid recruitment pressure, keep contact details separate and make clear that participation has no effect on grades or services. No payment is currently budgeted. A likely student-heavy sample will be acknowledged rather than described as representative of all newcomers.

Participant Information and Consent

Participants will receive a Chinese or English information sheet before scheduling and again at the session. It will explain the purpose, walking task, optional brief verbalisation, video and audio choices, data handling, withdrawal and contacts. Written or electronic consent will precede collection. Consent for recording video, recording audio, using de-identified quotations and displaying any identifiable image will be separate; agreement to research recording will not imply agreement to public display. The pilot also requires prior ethics clearance.

Participants may decline questions or recordings, pause or stop without penalty. We propose a seven-day window after each session for requesting removal of identifiable data; irreversible anonymisation will occur only after that window, subject to the approved protocol. Recruitment materials will contain a confirmed student contact and supervisor details. Unsafe weather, crowding or discomfort will lead to a pause, relocation or rescheduling; no research conversation will be required while crossing roads.

Confidentiality and Data Management

Participants will receive codes such as P01. Consent and contact records will be kept apart from clips, transcripts and matrices in institution-approved storage with access restricted to the team and authorised supervisors. The camera will be aimed at the relevant route context and actions, avoiding faces, private screens, messages and payment details wherever possible. Researchers will comply with venue recording restrictions. Incidental bystanders or private information will be excluded from analysis and removed or obscured in retained extracts; raw street footage will not be described as anonymous. Reports will use de-identified descriptions or redacted stills rather than identifiable raw video.

Audio and video will be used for recall and analysis, indexed by participant code and timestamp, and deleted under the approved retention plan. Storage location, retention period and contact details will be confirmed with Dr. Rui He before recruitment. Unneeded identifiable footage will be removed promptly under that plan. Raw participant media will not be uploaded to public AI services or personal automatic cloud backups; approved transcription tools, if any, will be disclosed. Participants may ask to pause recording and flag a clip for removal within the stated withdrawal window.

Anticipated Data Types

- Eligibility and contextual profiles, including time in Hong Kong, route familiarity and navigation habits.
- Timestamped field notes and whole-journey event timelines, including smooth episodes and interruptions.
- Consented key-episode videos, event indexes, contemporaneous comments, interview audio or notes and checked transcripts.
- Descriptive counts and approximate timings, coded case matrices and analytic/reflexive memos.
- An evidence-linked information classification, design propositions and illustrated boundary cases.

Research Timeline

The plan follows the 2026 course brief and its five presentations. Dates below are planning targets except the published course milestones. Participant activities remain conditional on prior ethics clearance. If clearance is delayed, we will present the prepared workflow and document the delay instead of collecting unapproved data.

Date or period	Course milestone	Planned work and evidence
19–24 September	Presentation 1 on 24 September	Finalise topic, literature map, questions, method rationale and ethics submission materials; no empirical findings claimed.
25 September–7 October	1 October has no class; data gathering is scheduled to begin	Seek clearance, scout route and rehearse as researchers. After clearance, complete one pilot and, if feasible, two main sessions.
8 October	Presentation 2	Show research workflow, protocol revisions and available pilot/main-study documentation; state actual approval and recruitment status.
9–28 October	Tutorial 15 October; analysis lecture 22 October	Complete the remaining main sessions toward six to eight in total, normally one or two per week. Transcribe, summarise and code alongside collection.

Date or period	Course milestone	Planned work and evidence
29 October	Presentation 3	Show updated workflow, data inventory, inclusion/exclusion decisions and examples of filtering and coding. Aim to close core recruitment.
30 October–12 November	Synthesis lecture 5 November; Presentation 4 on 12 November	Build the cross-case matrix; develop and challenge categories; present emerging synthesis, uncertainty and counterexamples.
13–26 November	Findings tutorial 19 November; Presentation 5 on 26 November	Refine evidence-linked design propositions and limitations; demonstrate the framework with cases; prepare a substantial report draft.
27 November–6 December	Individual written assessment 3 December; final report due 6 December	Protect individual assessment preparation time; revise and verify the 5,000–8,000-word English report, APA 7 references and required declarations; submit PDF under 10 MB with plagiarism report appended.

Workload and division of work

The core plan comprises one pilot, six to eight integrated sessions and one analysis cycle. Four students will work in two pairs, each responsible for three core participants and, if justified, one additional participant. Pair members will rotate observation and interview roles and cross-check an example from the other pair. Recruitment, transcription and analysis will proceed in small batches to avoid a backlog.

The planning basis remains three hours per student per week, or twelve team person-hours. Across eleven weeks, the budget is 132 person-hours: 16 for literature/source checking; 14 for protocol, recruitment, route and pilot; 22 for field sessions, travel and immediate notes; 28 for clip indexing, selective video review, transcription and coding; 12 for synthesis; 14 for presentations; 20 for the report; and 6 contingency hours. This equals 33 hours per student and excludes classes and individual assessment preparation. Six hours formerly reserved as contingency are allocated to media handling. The pilot will check this estimate; detailed coding will focus on three to five episodes per case. Optional additional participants will be dropped before increasing the agreed workload.

If capacity is lower, we will prioritise six well-documented cases, one route and a narrower set of information decisions. Videos are evidence for selected episodes, not a requirement for frame-by-frame coding of the whole walk. The information-card activity, large surveys, focus groups, longitudinal diaries and prototype comparisons are outside the core plan. Recruitment, observation, clip indexing and analysis will proceed in small batches to avoid a backlog.

Expected Outcome

We expect to describe how information and contextual demands relate to reported overwhelm, reassurance and manageable effort during unfamiliar walking navigation. A contextual classification and preliminary design framework will connect information role with timing, detail, delivery and support for confirmation or recovery. Case narratives will also describe participants' perceived implications for enjoying an outing, confidence and willingness to continue, including situations where extra information helps.

The contribution is a structured way to reason about navigation information design. The framework's completeness, usability and effectiveness in reducing burden require further evaluation. The small convenience sample, assigned route, camera and observer presence, optional verbalisation, retrospective recall and variation between apps constrain interpretation. A brief walk and one interview cannot demonstrate long-term withdrawal from urban exploration or prove that navigation caused fatigue or reduced enjoyment. The study cannot isolate newcomer status, infer neural processes or generalise to driving, cycling or all Hong Kong newcomers.

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